

**HI5 EDUCATION CENTRE
VICTORIOUS PRIMARY SCHOOLS**

PRIMARY SIX

MATHEMATICS

LESSON NOTES

TERM ONE, 2026

LESSON : 1

TOPIC : SET CONCEPTS

SUBTOPIC : EQUAL AND EQUIVALENT SETS

CONTENT :

Equal Sets

Equal sets are sets with the same number of elements of the same type. The symbol $=$ is used to denote equal sets.

Example:

1. If set $R = \{r, a, t\}$ and set $P = \{t, a, r\}$

$n(R) = 3$ members, $n(S) = 3$ members.

Members of R and S are similar

Sets R and P are therefore equal sets.

So we write; **$R = P$**

N.B: The arrangement of members does not matter provided they are exactly the same.

Equivalent Sets

Equivalent sets are sets with the same number of elements. The members may be different or the same. The symbol for equivalent is $\sim$.

Examples:

Set $B = \{4, 5, 6, 7, 8\}$ and Set $C = \{a, b, c, d, e\}$

$n(B) = 5$ members $n(C) = 5$ members

Therefore Sets B and C are equivalent since they both have 5 members each.

They can be written as **$B \sim C$**

ACTIVITY

- a) Define equal sets.
- b) What are equivalent sets?
- c) Given the sets below;

Set A = {0, 2, 4, 6, 8}

Set B = {2, 4, 6, 8, 10, 12, 14}

Set C = {s, n, a, i, l}

Set D = {4, 6, 8, 0, 2}

Set E = {n, a, i, l, s}

Set F is of even numbers between 1 and 15.

Use ‘**equal**’ or ‘**equivalent**’

- a) Set A and Set D
- b) Sets A and C
- c) Sets B and F
- d) Sets E and C
- e) Sets D and E
- d) Mr. Mulindwa has goats, cows and sheep on his farm and Mr. Muwonge has sheep, cows and pigs on

his farm. Write the sets of the two farms and state either they are equal or equivalent.

REFERENCES

MK MTC Pupil’s book 6 page 1

MK MTC Teachers’ book 6 page 1

Functional Primary Maths Pupil’s book 6 page 1–2

LESSON :

TOPIC : SET CONCEPTS

SUBTOPIC : UNEQUAL SETS

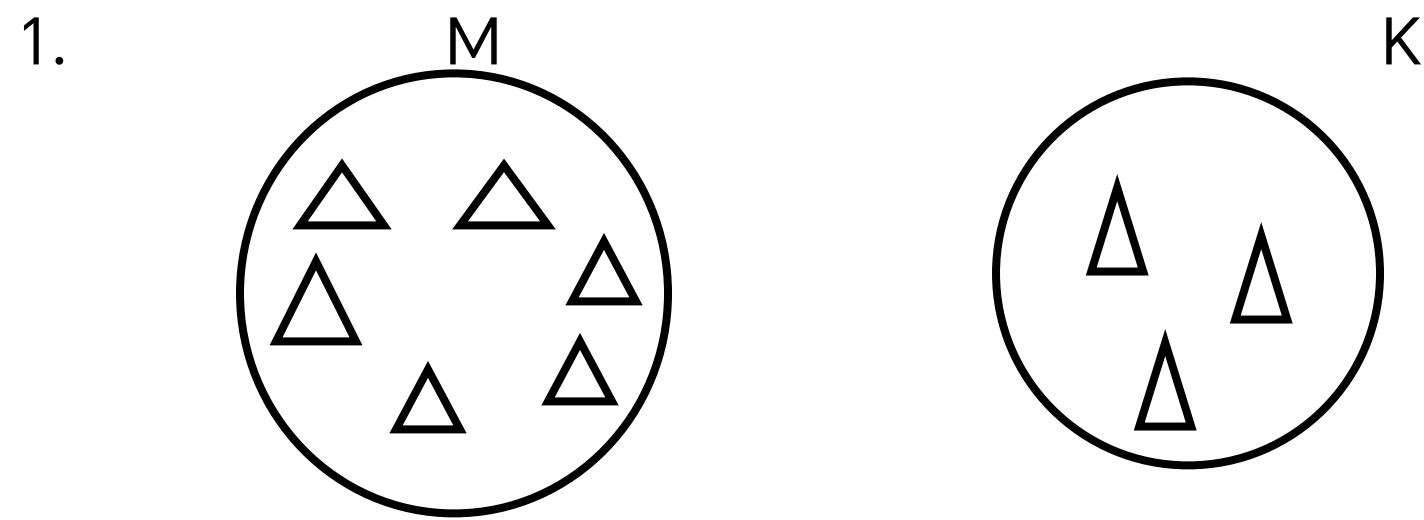
CONTENT:

Unequal sets are the sets with different members or different number of members.

N.B: Unequal means not equal.

The symbol for Unequal sets is $\neq$

Examples:



Set M has 6 members and Set K has 3 members.

Therefore Sets M and K are Unequal sets

$$M \neq K$$

2. Set T = {4, 5, 6, 7} and Set R = {a, p, q, k}

Set T is a set of 4 numbers and Set R is a set of 4 letters.

Sets T and R are unequal sets because their members are different though they have the same number of elements.

$$T \neq R$$

ACTIVITY:

Given the sets below, write equal or unequal.

1. Set P = {0, 2, 4, 6, 8} and Set Q = {8, 2, 4, 6}

Sets P and Q are _____ sets

2. Set B = {man, woman, boy}

Set C = {man, woman, girl}

Sets B and C are _____ sets.

3. Set D is a set of all the months of the year that start with letter J

Set E = {January, June, July}

Sets E and D are _____ sets

4. Given that sets $F = \{ \text{giraffe}, \text{rooster}, \text{hat} \}$ $G = \{ \text{circle}, \text{square}, \text{triangle} \}$

Sets F and G are _____ sets

REFERENCES

LESSON :

TOPIC : SET CONCEPTS

SUBTOPIC : INTERSECTION AND UNION SETS

CONTENT :

Intersection set

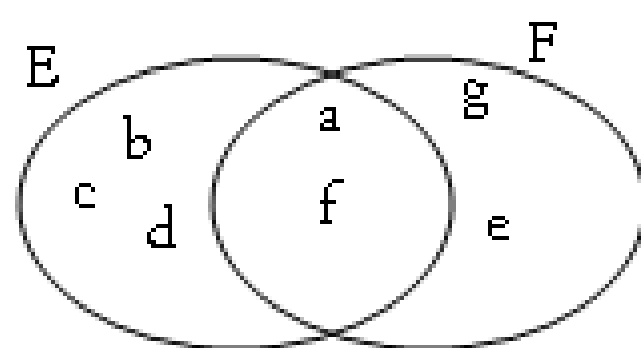
This is a set of common members of given sets.

Union set:

A set of all members in the given sets altogether.

Examples:

Given the venn diagram below,



1. Find $E \cap F$

$$\underline{E \cap F = \{a, f\}}$$

b) Find $n(E \cap F) = 2$ members

$$E \cap F = \{a, f\}$$

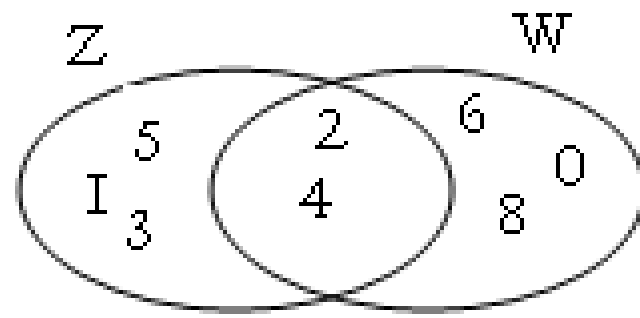
$$\underline{\text{Hence } n(E \cap F) = 2 \text{ members}}$$

c) What is $E \cup F$?

$$E \cup F = \{a, f, b, c, d, g, e\}$$

2. Given that Set $Z = \{1, 2, 3, 4, 5\}$ and $W = \{0, 2, 4, 6, 8\}$.

a) Represent the sets on a Venn diagram.



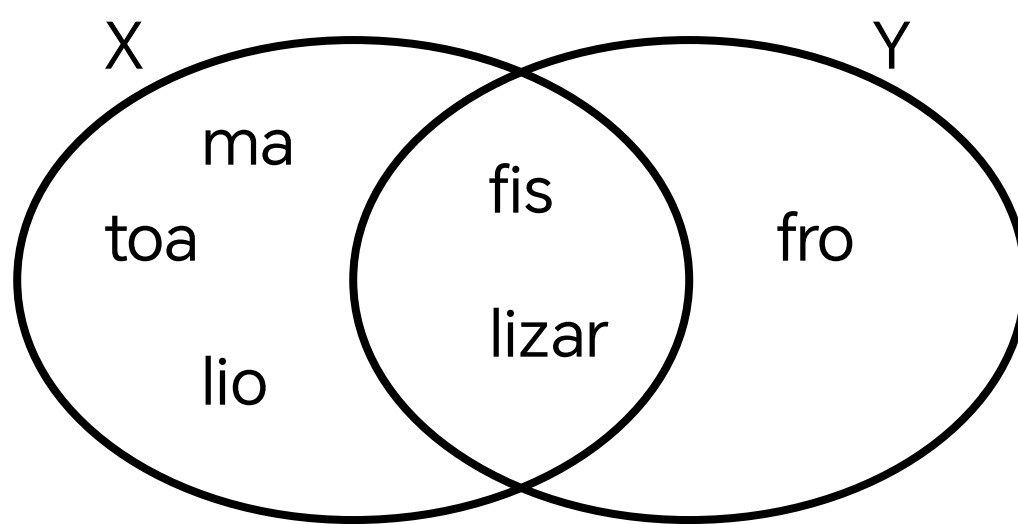
2. Find $n(Z \cup W)$

$$Z \cup W = \{0, 1, 2, 3, 4, 5, 6, 8\}$$

$$n(Z \cup W) = 8 \text{ members}$$

ACTIVITY:

a) Use the Venn diagram to answer the questions.



a) Find $X \cap Y$.

b) Find $n(X \cap Y)$.

c) Find $Y \cup X$.

d) What is $n(X \cup Y)$?

2. Given that Set K is a set of all factors of 12 and Set L is a set of all factors of 30.

i. Find $K \cap L$.

ii. Find the union set of K and L.

iii. How many elements are in $L \cup K$?

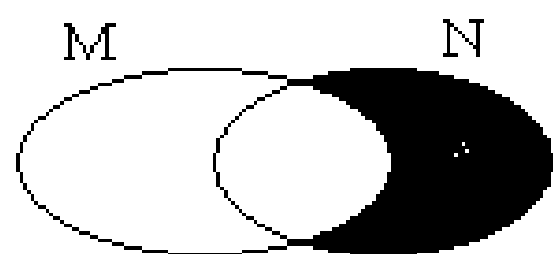
iv. Find $L \cup K$

REFERENCES

LESSON :
TOPIC : SET CONCEPT
SUB TOPIC : DIFFERENCE OF SETS
CONTENT :

Examples:

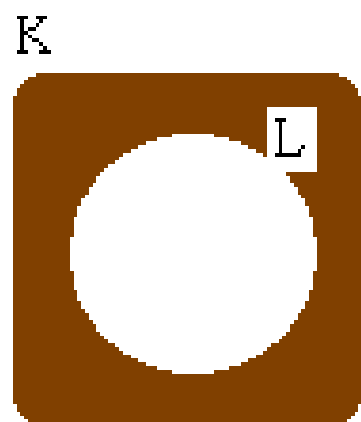
i. Shade $N - M$ on the Venn diagram below.



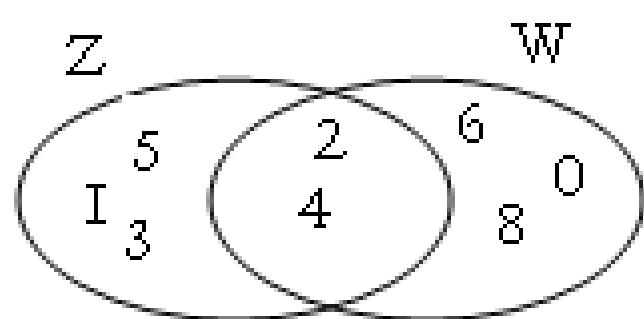
NB $N - M$ refers to the region for M only.

It also means: $N - (M \cap N)$

ii. Shade $K - L$ in the sets.



3. Given the Venn diagram below:-



1. Find $W - Z$

$W - Z = \{0, 6, 8\}$

b. Find $n(X - W)$

$Z - W = \{1, 3, 5\}$

$n(Z - W) = 3 \text{ members.}$

4. Given that Set R is a set of all vowel letters in the word “chair” and Set K is a set of all vowel letters in the word “education”.

1. Find $K - R$

2. Find $n(R - K)$

3. Set $R = \{a, i\}$

4. Set $K = \{a, e, i, o, u\}$

a) $K - R = \{e, o, u\}$

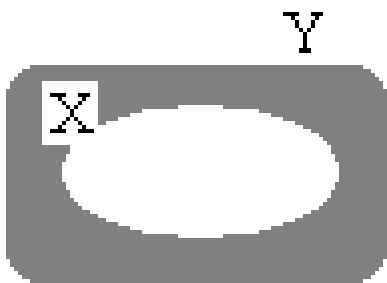
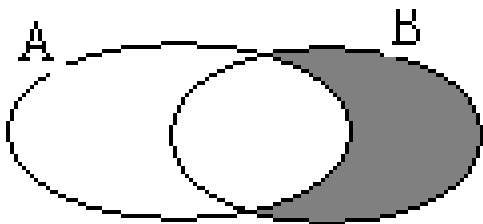
b) $n(R - K)$

$R - K = \{ \}$

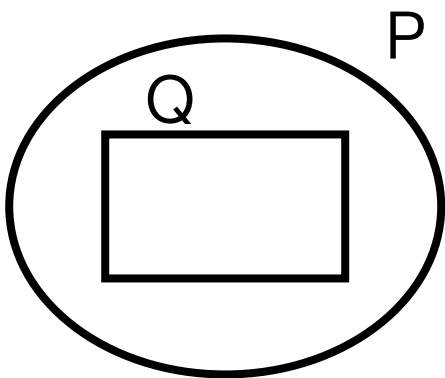
$n(R - K) = 0$

ACTIVITY:

1. Describe the shaded regions



2. Shade $P - Q$

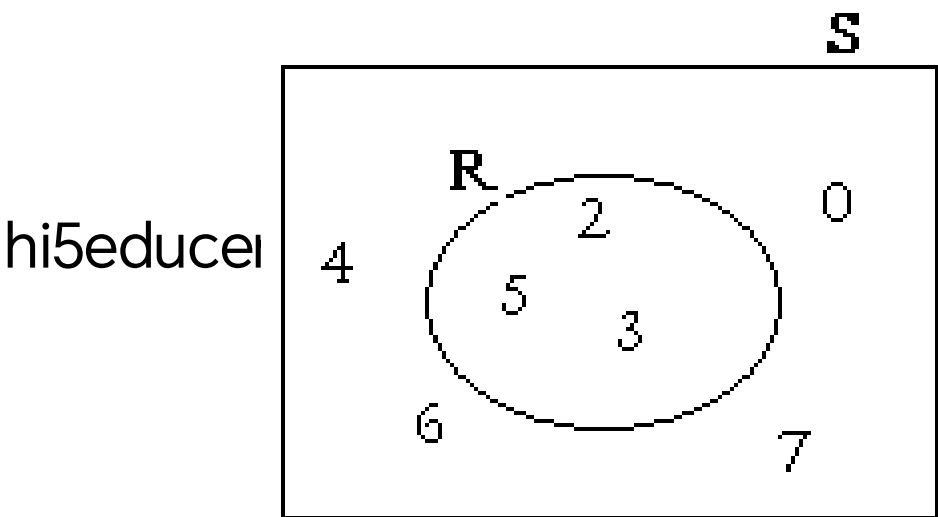


3. Set $B = \{a, h, k, r, s\}$ Set $H = \{b, h, t, r, v\}$.

i. Find i) $H - B$

ii. $n(B - H)$

4. Study the diagram and answer the questions



i. List the members of set R

5. Set T is a set of all multiples of 4 less than 19. Set M is a set of all factors of 24.
- Find $T - M$
 - Find $n(M - T)$

REFERENCES:

MK Maths Pupil's book 6 page 11-12

MK Maths Teachers' book 6 page 8-9

LESSON :

TOPIC : SET CONCEPTS

SUBTOPIC : COMPLEMENT OF SETS

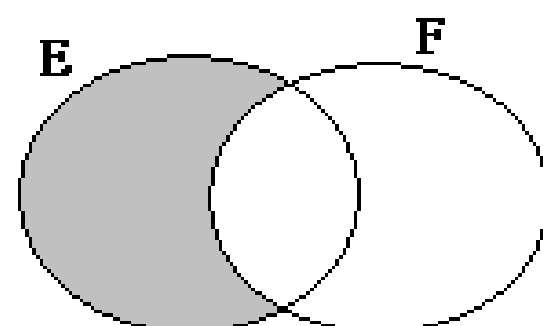
CONTENT :

Complement of a set refers to the region or members with in the union of the given sets but do not belong to that given set.

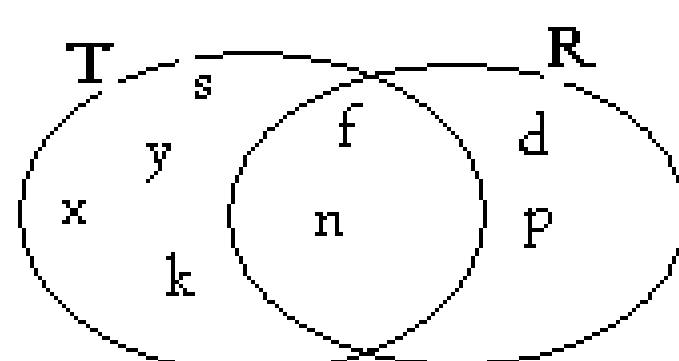
We use the apostrophe sign to write the complement of a set e.g. the complement of a set B is written as B' .

Example.

1. Shade F' in the sets.



2. Given the sets below, find R'



$$R^I = \{k, s, x, y\}$$

3. Set $Z = \{p, q, r, s, t\}$ Set $Y = \{r, s, t, u, v\}$

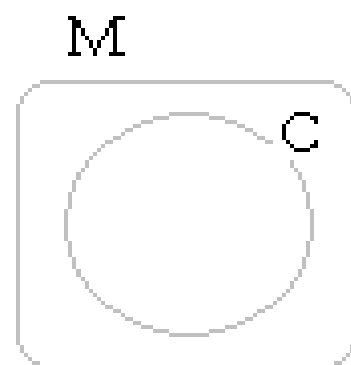
Find $n(Z')$

$$Z' = \{u, v\}$$

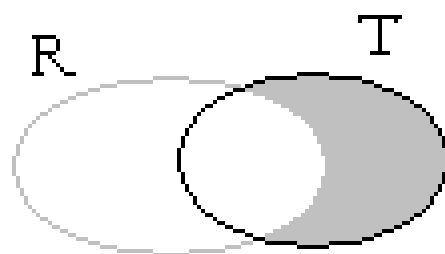
$$\underline{n(Z') = 2 \text{ members.}}$$

ACTIVITY:

1. Shade the $M - K$ on the Venn diagram below



2. Describe the shaded region in terms of complement



3. Set W is a set of all composite numbers less than 10 and set X is a set of all even numbers less than 16. Find $n(X')$

REFERENCES

MK Maths Pupil's book 6 page 9-10

MK Maths Teachers' book 6 page 5-7

Functional Primary Maths Pupil's book 6 page 4-5

LESSON :

TOPIC : SET CONCEPTS

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SUBTOPIC : UINVERSAL AND SUBSETS

CONTENT :

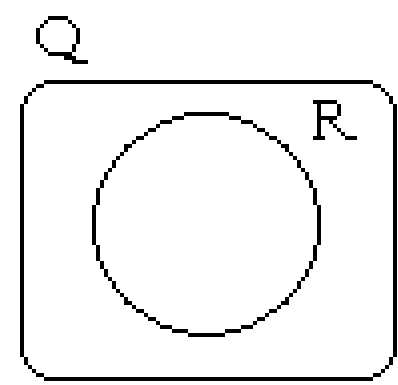
Universal set

This is the mother set or the main/ bigger set. For example, if set Q is a set of all children in Victorious Primary School and set R is a set of all children in P.6 class of Victorious, then, set Q is a universal set

Note; Set R is just part of Set Q.

The symbol for universal set is $\mathcal{E}$

The sets Q and R can be represented as



Subset:

A subset is the smaller set which can be obtained from any given set. For example set R above is a subset of set Q since it is just part of Q. The symbol $\subset$ is used to imply 'is a subset of'

The sets above can be written as; $R \subset Q$

Proper subsets

Proper subsets are subsets with the exception of the main set itself.

Number of proper subsets is got by using $(2^n) - 1$

Consider

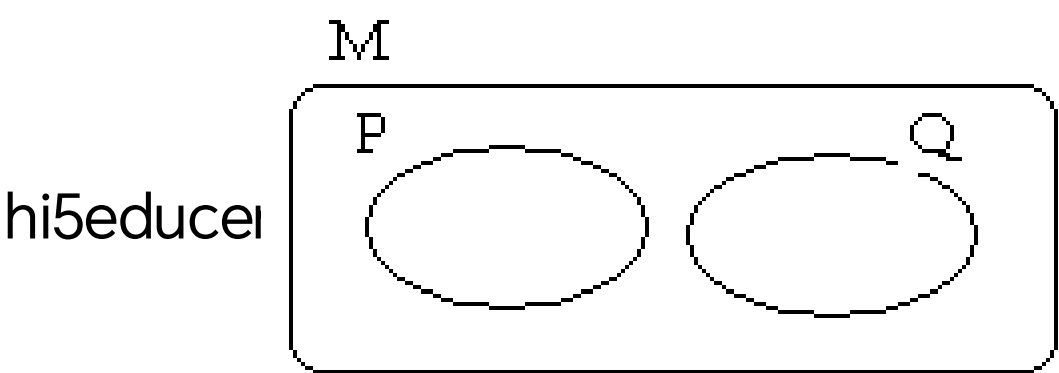
Given that,

Set M is a set of all farmers in Masiku Village.

Set P is a set of farmers who grow food crops.

Set Q is a set of farmers who rear animals.

Represent the sets on a venn diagram



N.B: - All farmers (M) is the universal set.

All farmers who grow food crops (P) is a subset of M, thus $P \subset M$

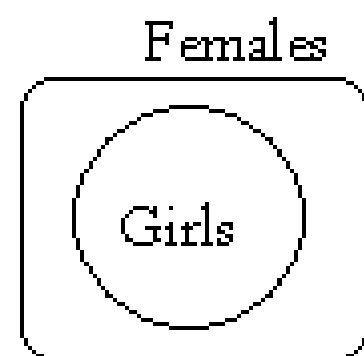
All farmers who rear animals (Q) is subset of M, thus $Q \subset M$

Farmers who rear animals and grow food crops is a subset of M, thus $(P \cup Q) \subset M$

Farmers who grow other crops is a subset of M, thus $(P \cup Q)' \subset M$

ACTIVITY:

1. Write the relationship between the sets in the Venn diagram below



2. Draw a Venn diagram to show that all animals (A) are Living things (L).
3. It is true that Kampala (K) is found in Uganda (U) which is in Africa (A).
Represent this statement on a venn diagram.

REFERENCES

MK Maths Teachers' book 6 page 5-6

MK Maths Pupil's book 6 page 3-14

Functional Primary Maths Pupil's book 6 page 8-9

LESSON :

TOPIC : SET CONCEPTS

SUBTOPIC : LISTING AND FINDING NUMBER OF SUBSETS

CONTENT :

Subsets are smaller sets obtained from a given set.

Listing subsets:

Example

1. Set $B = \{2, 4, 6\}$. List all the subsets in set B.

$\{ \}, \{2\}, \{4\}, \{6\}, \{2, 4\}, \{2, 6\}, \{4, 6\}, \{2, 4, 6\}$.

Note:

The empty set is also a subset of any given set. The set itself is also a subset of itself.

Finding Number of Subsets

This can be done in either of two ways:

5 First listing the subsets then count them and finally state the number of subsets formed or

6 Using the formula; thus,

Number of Subsets = 2^n

Where n stands for number of elements in that given set.

Examples

1. Set $P = \{a, b, c, d\}$. Find the number of subsets in set P

$n(P) = 4$ members

No. of subsets = 2^n

$= 2^4$

$= 2 \times 2 \times 2 \times 2$

$= 16$ Subsets.

P has 16 subsets.

2. Given that $n(K) = 6$. Find the number of subset in set K.

$$n(K) = 6$$

$$\text{No. of subsets} = 2^n$$

$$= 2^6$$

$$= 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

$$= 64 \text{ Subsets.}$$

K has 64 subsets.

ACTIVITY:

1. List all the subsets in each of the given sets
 - a) Set $B = \{2, 3, 4\}$
 - b) Set $M = \{a, b, c, d\}$
2. Find the number of subsets in each set by first listing them.
 - a) Set $Z = \{p, q, r\}$
 - b) Set $R = \{6\}$
3. Using the formula, calculate the number of subsets the sets below:
 - c) Set $W = \{0, 3, 6\}$
 - d) Set $T = \{ \}$
 - e) Set X is a set of 3 blue cows on Mr. Muwonge's farm. Calculate the number of subsets in set X .

REFERENCES

MK Maths Pupil's book 6 page 5-7

MK Maths Teachers' book 6 page 3-5

Functional Primary Maths Pupil's book 6 page 8-9

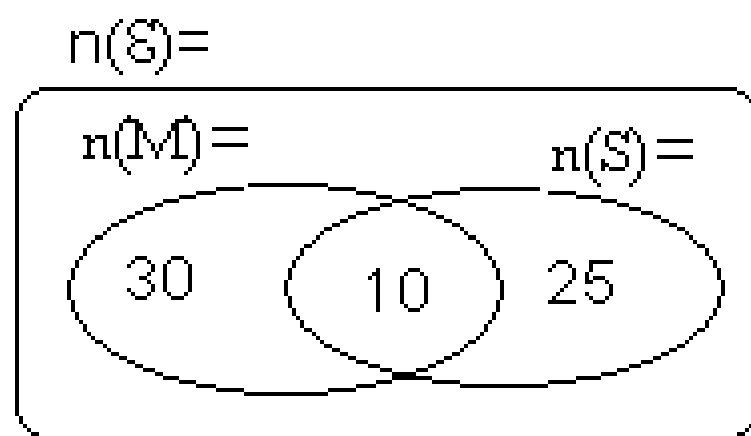
LESSON :

TOPIC : SET CONCEPTS

SUBTOPIC : APPLICATION OF VENNDIAGRAMS

CONTENT :

The Venn diagram below shows how all P.6 children prefer two clubs i.e Maths club and Science club.



a) How many pupils prefer Maths club?

$$\text{Maths club} = n(M) \text{ only} + n(M \cap S)$$

$$= 30 + 10$$

$$= \underline{40 \text{ pupils.}}$$

b) How many pupils prefer both clubs?

$$\underline{10 \text{ pupils prefer both clubs}}$$

c) How many pupils prefer only one club?

$$n(M) \text{ only} + n(S) \text{ only}$$

$$= 30 + 25$$

$$= \underline{55 \text{ pupils}}$$

d) How many pupils are in P. 6 class?

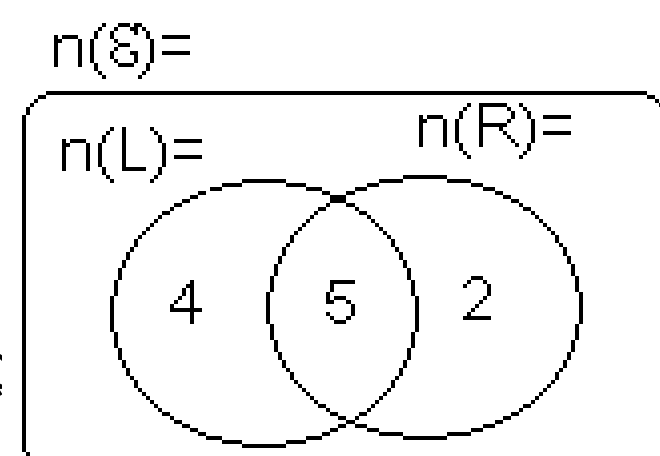
$$n(E) = n(M) \text{ only} + n(M \cap S) + n(S) \text{ only}$$

$$30 + 10 + 25$$

$$= \underline{65 \text{ pupils}}$$

ACTIVITY:

The Venn diagram below shows how a school football team some use left leg (L), others use the right leg (R) and few use both legs



- How many members are in the school team?
- Find the number of players who use left leg.
- How many players use only one leg?
- How many players use either left or right leg?
- How many players use at least one leg?

REFERENCES

MK Maths Pupil's book 6 page 29–30

MK Maths Teachers' book 6 page 10

Functional Primary Maths Pupil's book 6 page 10–12

LESSON: 10

TOPIC: SET CONCEPTS

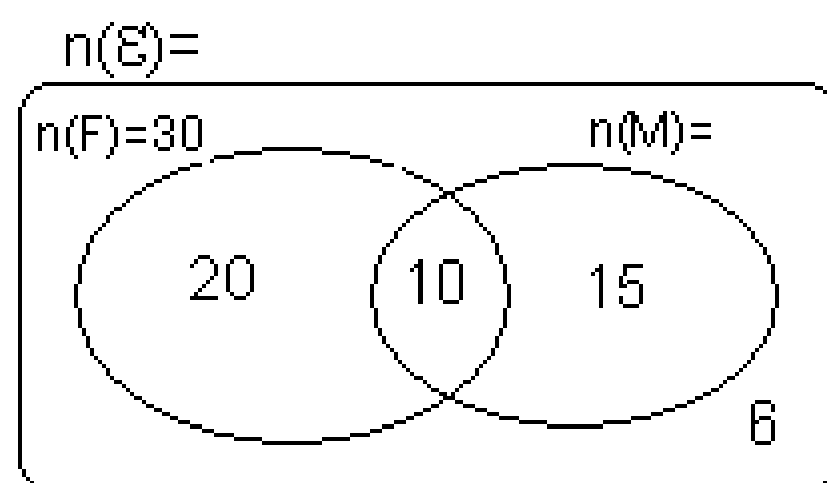
SUBTOPIC: APPLICATION OF SETS

CONTENT:

Representing information on a Venn diagram

- In a p.6 class, 30 pupils prefer fish (F), 15 prefer meat (M) only, 10 prefer both fish and meat and 6 don't like any of the two.
 - Draw a Venn diagram to show the information.

$$n(F)=30 \quad n(M)_{\text{only}}=15 \quad n(F \cap M)=10 \quad n(F \cup M)=26, n(E)=6$$



b) How many pupils prefer fish only?

$$\begin{aligned}n(F)\text{only} &= n(F) - n(F \cap M) \\&= 30 - 10 \\&= \underline{20 \text{ pupils}}\end{aligned}$$

c) What is the population of this class?

$$\begin{aligned}n(\mathcal{E}) &= n(F) + n(F \cap M) + n(M)\text{only} + n(F \cup M)^1 \\&= (30 - 10) + 15 + 10 + 6 \\&= 20 + 25 + 6 \\&= 51 \text{ pupils}\end{aligned}$$

d) Find the probability of picking at random a member likes fish only

$$n(F) = 30 \quad n(\mathcal{E}) = 51$$

$$\begin{aligned}P(\text{Fish}) &= \frac{n(F)}{n(\mathcal{E})} \\&= \frac{30}{51}\end{aligned}$$

ACTIVITY:

1. In a family, 12 members use English (E), 8 use Luganda (L), 4 use both English and Luganda and 3 use neither of the two languages.
 - a) Draw a venn diagram to represent the information
 - b) B) How many members use only one language?
 - c) If each member in this family was given sh. 10,000 for weekend, how much money was given to this family?
 - d) What is the probability of getting a member who uses English only?

REFERENCES

MK Maths Pupil's book 6 page 22–24

MK Maths Teachers' book 6 page 11–12

LESSON : 3 PERIODS

TOPIC : SET CONCEPTS

SUBTOPIC : APPLICATION OF SETS

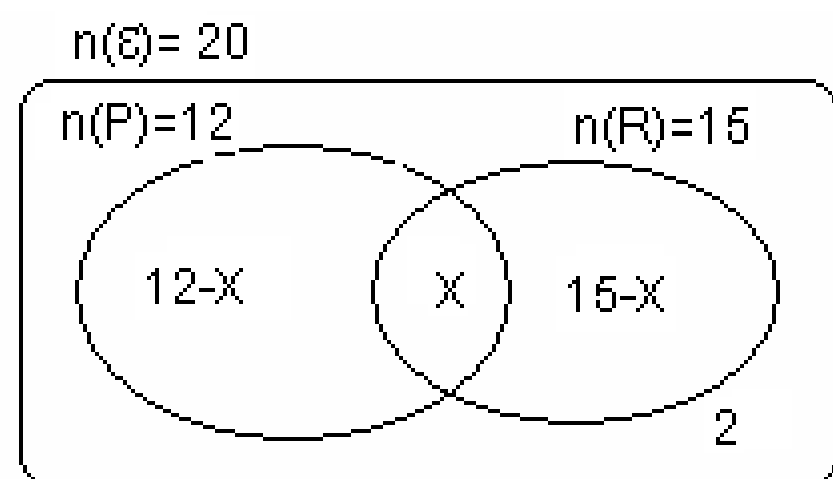
CONTENT :

- a) In a school of 20 teachers, 12 teachers prefer posho (P), 15 prefer rice (R), some prefer both posho and rice and 2 prefer neither of the two kinds of food.

1. Represent the information on a venn diagram

Let $n(P \cap R)$ be X

$$n(P)=12 \quad n(R)=15 \quad n(P \cap R)=X \quad n(P \cup R)^1=2, n(\mathcal{E})=20$$



2. Find the number of teachers who prefer both posho and rice.

Note; for both, it is represented by X

Method I:

$$X + 12 - X + 15 - X + 2 = 20$$

$$X - X - X + 12 + 15 + 2 = 20$$

$$-X + 29 = 20$$

$$-X + 29 - 29 = 20 - 29$$

$$-X = -9$$

$$X = 9$$

Method III:

$$15 + 12 - X + 2 = 20$$

$$27 - X + Y + 2 = 20 + X$$

$$27 + 2 = 20 + X$$

$$29 - 20 = 20 - 20 + X$$

$$9 = X$$

$$\underline{X = 9}$$

Therefore 9 teachers prefer both posho and rice.

Method II:

$$12 + 15 - X + 2 = 20$$

Method IV:

$$X = (15 + 12 + 2) - 20$$

$$X = 29 - 20$$

$$\underline{X = 9}$$

$$12 + 15 - X + X + 2 = 20 + X$$

$$27 + 2 = 20 + X$$

$$29 - 20 = 20 - 20 + X$$

$$9 = X$$

$$\underline{X = 9}$$

3. Find the number of teachers who prefer only one type of food.

$$(12 - X) + (15 - X)$$

$$(12 - 9) + (15 - 9)$$

$$3 + 6$$

$$\underline{= 9 \text{ teachers}}$$

ACTIVITY:

In a class of 40, 25 pupils prefer Maths (M), 20 prefer English (E), 10 prefer both Maths and English and 5 prefer neither Maths nor English.

- Represent the information on a Venn diagram
- Find the number of pupils who like both Maths and English
- How many pupils prefer only one subject?
- What is the probability of picking a pupil at random who prefers Maths only to be the class monitor?

REFERENCES

MK Maths Pupil's book 6 page 23-25

MK Maths Teachers' book 6 page 13-14

LESSON :

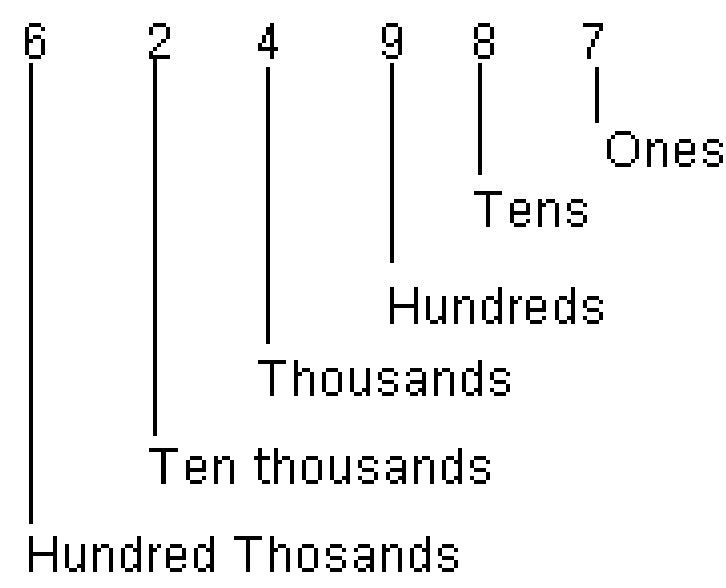
TOPIC : NUMERATION SYSTEMS AND PLACE VALUE

SUBTOPIC : PLACE VALUES AND VALUES

CONTENT :

Examples :

2 Find the place value of each digit in 6 2 4 9 8 7



3 Find the value of each digit in 86427.

T Th	Th	H	T	O
8	6	4	2	7

Value of a digit = digit x place value

Value of:

$7 = 7 \times 1 = 7$		$6 = 6 \times 1000 = 6000$
$2 = 2 \times 10 = 20$		$8 = 8 \times 10000 = 80000$
$4 = 4 \times 100 = 400$		

4 State the place value of each digit in 634.78

H	T	O	Tth	Hth
6	3	4	.7	8

Place value of: 6 = Hundreds

3 = Tens

4 = Ones

7 = Tenth

8 = Hundredths

5 Find the value of each digit in 72.929

T	O	Tth	Hth	THth
7	2	.9	2	9

Value = digit x place value

$$7 \times 10 = 70$$

$$9 \times \frac{1}{10}$$

$$= \frac{9}{10}$$

$$= 0.9$$

$$2 \times 1 = 2$$

$$2 \times \frac{1}{100}$$

$$= \frac{2}{100}$$

$$= 0.02$$

$$9 \times \frac{1}{1000}$$

$$= \frac{9}{1000}$$

$$= 0.009$$

ACTIVITY:

a) Write the place value of each digit

1. 369853

1. Find the value of each of the digits.

2. 1900624

1. 66.42

3. 783.36

1. 1986.797

4. 89.8663

1. 616.789

b) Find the sum of the place value of 7 and the value of 2 in 200763

c) What is the product of the value of 8 and place value of 4 in 863.47?

REFERENCES

MK Maths Pupil's book 6 page 34-35

MK Maths Teachers' book 6 page 31-33

Functional Primary Maths Pupil's book 6 page 19-25

LESSON :

TOPIC : NUMERATION SYSTEMS AND PLACE VALUE

SUBTOPIC : EXPANDING NUMBERS

CONTENT :

a) Expand 4963 in place value form

TH	H	T	O
4	9	6	3

$$= 4 \times 1000 + 9 \times 100 + 6 \times 10 + 3 \times 1$$

b) Expand 68.604 in place value form

T	O	T	H	TH
6	8.	6	0	4

$$= (6 \times 10) + (8 \times 1) + (6 \times 0.1) + (0 \times 0.01) + (4 \times 0.001)$$

$$= (6 \times 10) + (8 \times 1) + (6 \times 0.1) + (4 \times 0.001)$$

OR

$$= (6 \times 10) + (8 \times 1) + (6 \times 1/10) + (4 \times 1/1000)$$

c) Expand 68.604 in value form.

T	O	Tth	Hth	THth
6	8.	6	0	4

$$= (6 \times 10) + (8 \times 1) + (6 \times 1/10) + (4 \times 1/1000)$$

$$= (6 \times 10) + (8 \times 1) + (6 \times 0.1) + (0 \times 0.01) + (4 \times 0.001)$$

$$= (6 \times 10) + (8 \times 1) + (6 \times 0.1) + (4 \times 0.001)$$

$$\underline{\quad\quad\quad = 60 + 8 + 0.6 + 0.00 + 0.004 \quad\quad\quad}$$

d) Expand 85.764 in power form

Note: In expanding using powers/exponents, the whole numbers take positive powers while the decimal places take powers. These exponents/powers are often.

ACTIVITY:

a) Expand the following using powers of ten.

a) 68849

b) 2.665

c) 1963.304

b) Expand the following in place value form

1. 1717

2. 634.578

3. 49.857

c) Expand the following in value form

1. 54321

2. 78.902

REFERENCES

MK Maths Pupil's book 6 page 34–35

MK Maths Teachers' book 6 page 31–33

Functional Primary Maths Pupil's book 6 page 19–25

LESSON :

TOPIC : NUMERATION SYSTEMS AND PLACE VALUE

SUBTOPIC : WRITING EXPANDED NUMBER IN SINGLE NUMBER

CONTENT :

1. What number has been expanded below?

$$60000 + 20 + 500 + 3$$

$$60000$$

$$500$$

$$20$$

$$\begin{array}{r} + \quad 3 \\ \hline \end{array}$$

$$\begin{array}{r} \mathbf{60523} \\ \hline \end{array}$$

2. Find the number that has been expanded

$$(7 \times 1000) + (6 \times 0.1) + (5 \times 10)$$

$$= 7000 + 0.6 + 50$$

$$7000.0$$

$$50.0$$

$$\begin{array}{r} + \quad 0.6 \\ \hline \end{array}$$

$$\begin{array}{r} \mathbf{7050.6} \\ \hline \end{array}$$

3. Namuli expanded a certain number and got,

$$(6 \times 10^4) + (5 \times 10^0) + (3 \times 10^1) + (7 \times 10^3)$$

What number did she expand?

ACTIVITY:

Find the numbers which have been expanded below

- a) $6000 + 20 + 7$
- b) $(7 \times 1000) + (8 \times 10) + (9 \times 100) + (7 \times 1)$

REFERENCES

MK Maths Pupil’s book 6 page 36–37

MK Maths Teachers’ book 6 page 36–37

Functional Primary Maths Pupil’s book 6 page 22–24

LESSON :

TOPIC : NUMERATION SYSTEMS AND PLACE VALUE

SUBTOPIC : WRITING NUMBERS IN WORDS

CONTENT :

- 1. Write 6 2 4 9 1 4 in words

Thousands			Units		
H	T	O	H	T	O
6	2	4	9	1	4

624,914 = Six hundred twenty four thousand nine hundred fourteen

- 2. Write 1 9 0 0 3 0 0 4 7 in words

Millions			Thousands			units		
H	T	O	H	T	O	H	T	O
1	9	0	0	3	0	0	4	7

190,030,047 = One hundred ninety million thirty thousands forty seven

3. Write 24.63 in words

T	O		T th	H th
2	4	.	6	3

24.63 = Twenty four and sixty three

hundredths

ACTIVITY:

Write the following in words

1. 62,493

2. 171717

3. 9009009

4. 666666666

5. 100100100
6. 14.14

7. 272.009

8. 4634.665

9. 0.0004

10. 6.789

REFERENCES

- MK Maths Pupil’s book 6 page 39
- MK Maths Teachers’ book 6 page 39–40
- Functional Primary Maths Pupil’s book 6 page 24–26

LESSON :
TOPIC : NUMERATION SYSTEMS AND PLACE VALUE
SUBTOPIC : WRITING numbers from words to figures
CONTENT :
a) Write in figures

Seventy four million, six hundred ninety two thousand, five hundred eleven.

Millions			Thousands			Units		
H	T	O	H	T	O	H	T	O
0	7	4	6	9	2	5	1	1

= 74,692,511

b) Seventy nine point four five six.

Seventy nine point four five six = 79.456

c) Our hundred nine and forty six hundredths

One hundred nine = 109

Forty six hundredths = $\frac{46}{100}$ = 0.46

109.00
 0.46
109.46

ACTIVITY:

- 1. Write the following in figures
 - i) Seventeen million, seven thousand, seventeen
 - ii) To hundred thousand, three hundred sixty four
 - iii) Sixty six point seventy six million, five hundred forty three thousand, two hundred ten.
 - iv) Ninety and nine thousandths
- 2. Write the number represented on the abacus

REFERENCES:

MK Maths Pupil’s book 6 page 40–46
MK Maths Teachers’ book 6 page 40–45
Functional Primary Maths Pupil’s book 6 page 23–25

LESSON :

TOPIC : NUMERATION SYSTEMS AND PLACE VALUE

SUBTOPIC : ROUNDING OFF WHOLE AND DECIMAL NUMBERS

CONTENT :

Rounding off means – correcting to the nearest values

Other terms;

- Rounding up
- Rounding down

Examples:

1. Round off 4965 to the nearest,

1. Tens

Note: When the next number to the right of the required place is 5 and above, we round up (add one to the digit in the required place) and when it is less than 5, we round (do not add one to the digit) in the required place.

2. Round off 96329 to the nearest hundreds

			T th	T ^h	H	T	O
			9	6	3	2	9
+			0	0	0	0	0
			9	6	0	0	0

3. Round off 728.36 to the nearest whole number

4. Note: Rounding off to the nearest whole number means to the nearest ones.

5. Round off 68.964 to the nearest tenths.

ACTIVITY:

1. Round off 666 to the nearest tens
2. Round off 19634 to the nearest THOUSANDS
3. Round off 45.36 to the nearest tenths
4. WRITE 689.99 to the nearest whole number.
5. WRITE 999.9999 to the nearest thousandths
6. Round off 123121 to the nearest ten thousands.

REFERENCES

MK Maths Pupil's book 6 page 40–46

MK Maths Teachers' book 6 page 40–45

Functional Primary Maths Pupil's book 6 page 23–25

LESSON :

TOPIC : NUMERATION SYSTEMS AND PLACE VALUE

SUBTOPIC : HINDU ARABIC TO ROMAN NUMERALS

CONTENT :

Letters used in Roman Numerals;

I – 1	C – 100
V – 5	D – 500
X – 10	M – 1000
L – 50	

a) Write 462 in Roman Numerals

$$\begin{aligned}
 462 &= 400 + 60 + 2 \\
 &= CD + LX + II \\
 &= CDLXII
 \end{aligned}$$

b) Write 1629 in Roman Numerals

$$1629 = 1000 + 600 + 20 + 9$$

$$= M + DCCC + XL + IX$$

$$= MDCCCXLIX$$

Note: We use capital letter form when writing in Roman Numerals

ACTIVITY

1. Write the following in Roman Numerals

1. 49 d) 2424

2. 176 e) 964

3. 3332 f) 1234

2. Lukule was given 3965 books. Express his number in Roman Numerals.

3. How would a Roman girl write 5260?

REFERENCES

MK Maths Pupil's book 6 page 40-46

MK Maths Teachers' book 6 page 40-45

Functional Primary Maths Pupil's book 6 page 23-25

LESSON :

TOPIC : NUMERATION SYSTEMS AND PLACE VALUE

SUBTOPIC : ROMAN NUMERALS TO HINDU ARABIC

CONTENT :

1. Write CDLX in Hindu Arabic Numerals

Note: When a letter of less value comes before that of a greater value, it means subtraction

$$CDLX = CD + LX$$

$$= 400 + 60$$

$$= 460$$

2. Express CMLXVIII in Hindu Arabic Numerals

$$CMLXVIII = CM + LX + VIII$$

$$= 900 + 60 + 8$$

$$= 968$$

3. Wasswa wrote MMMDXIX on a card. What number is this in Hindu Arabic Numerals?

$$\text{MMMDXIX} = \text{MMM} + \text{D} + \text{XIX}$$

$$= 3000 + 500 + 19$$

$$= 3519$$

Activity

Write the following in Hindu Arabic Numerals

1. CCX

6. MMLXXIV

2. CCCIX

7. CCCIII

3. DCCCLXXXVIII

8. CDVII

4. CDXCII

9. XIX

5. LIX

10. CXIX

LESSON :

TOPIC : OPERATION ON NUMBERS

SUBTOPIC : ADDITION OF NUMBERS

CONTENT :

- a) Add: $469046 + 63942$

$$\begin{array}{r} 469046 \\ + 63942 \\ \hline \end{array}$$

$$\hline$$

$$\underline{532988}$$

- b) There are four million sixty thousand people in Eastern Uganda, six million forty thousand six hundred in Western Uganda and three million fifty thousand in Northern part. Find the total population of the three regions.

$$\begin{array}{rcl} \text{Eastern} & - & 4060000 \\ \text{Northern} & - & \underline{3050000} \\ \hline \end{array}$$

$$\hline$$

Total - 7 110 000

The total population in the regions is 7,110,000 people.

ACTIVITY

1. Add the following numbers
 1. $96114 + 3224$
 2. $630004 + 99963$
 3. $17171717 + 222222$
 4. $10000 + 100000 + 1000$
2. There are 46920 female and 32690 male in Kamuli District. Find the population of the district.
3. Uganda's population is approximately 33 million and that of South Africa is 66.5 million. Find the approximate total population

Reference

MK Maths Pupil's book 6 page 40–46

MK Maths Teachers' book 6 page 40–45

Functional Primary Maths Pupil's book 6 page 23–25

LESSON :

TOPIC : OPERATION ON NUMBERS

SUBTOPIC : subtraction of large numbers

CONTENT :

a) Subtract $85604 - 64503$

- Arrange the numbers vertically according to the place values of the given digits the subtract.

b) Subtract 2896475 from 8331843

Interpretation of the 'from' operation, thus

$$8331843 - 2896475$$

- arrange vertically and subtract

c) There are 49625 text books in Victorius Library. 16240 are maths books and the rest are other subjects. How many books are for other subjects?

Total no. of books - 49625

Number of maths - 16240

Other subjects - 33385

ACTIVITY:

1. Subtract the following

1. $40000 - 3000$

2. $562003 - 49999$

3. $634963241 - 100100100$

2. Subtract 99 from 10000000

3. What is the difference between 3694 and 76300?

4. How far is 50,000 metres away from 19500m?

5. In a country of 36 million people, 2,563,200 are adults and the rest are child. Find the number of children in this country.

LESSON :

TOPIC : OPERATION ON NUMBERS

SUBTOPIC : MULTIPLICATION

CONTENT :

1. Multiply 242×12

$$\begin{array}{r} 242 \\ \times 12 \\ \hline \end{array}$$

$$\begin{array}{r}
 484 \\
 + 2420 \\
 \hline
 2904 \\
 \hline
 \end{array}$$

2. A school bus carries 68 passengers when full. If it makes 42 trips, how many passengers will be carried altogether?

In one trip, it carries 68 passengers

In 42 trips it carries (42 x 68) passengers

$$\begin{array}{r}
 42 \\
 \times 68 \\
 \hline
 2520 \\
 + 336 \\
 \hline
 2856 \\
 \hline
 \end{array}$$

The bus carries 2856 passengers in the 42 trips

ACTIVITY:

Multiply the following

1. 66×424

4. 9103×133

2. 117×24

5. 817×1313

3. 6636×36

6. 312×495

The average number of children in 136 schools in Mukono district is 1250. Find the total population in all the schools.

Find the product of 396 and 3298.

Reference

MK Maths Pupil's book 6 page 40–46.

MK Maths Teachers' book 6 page 40–45.

Functional Primary Maths Pupil's book 6 page 23–25.

LESSON :

TOPIC : OPERATION ON NUMBERS

SUBTOPIC : DIVISION OF NUMBERS

CONTENT :

1. Divide 7620 by 20

$$\begin{array}{r} 0381 \\ 20 \overline{) 7620} \\ \underline{+ 0} \\ 76 \\ \underline{- 60} \\ 162 \\ \underline{- 160} \\ .20 \\ \underline{- 20} \end{array}$$

2. Divide $76050 \div 234$

Children be encouraged to divide using multiples if the divisor (see P.5 notes)

ACTIVITY:

Divide the following numbers:

2 $1256 \div 13$

3 $25610 \div 132$

4 Divide 5600 by 250

REFERENCES:

Functional Primary Maths Pupil's book 6 page 47–48

MK Maths Teachers' book 6 page 72–73

LESSON :

TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : DIVISIBILITY TESTS

CONTENT :

Divisibility test by 2

A number is divisible by 2 when it ends with, 0, 2, 4, 6, 8

e.g. 66, 200, 7204, 98, 24, 62

Divisibility test by 3

A number is divisible by 3 when the sum of its digits is 3 or 6 or 9.

e.g. i) 291

$$291 \quad 2+9+1 = 12$$

$$12 - 1+2 = 3$$

Therefore 291 is divisible by 3

ii) State whether 12631 is divisible by 3 or not.

$$12631 - 1+2+6+3+1$$

$$= 13$$

$$= 1+3$$

$$= 4$$

Therefore 12631 is divisible by 3

Divisibility test by 5

A number is divisible by 5 when it ends with either 0 or 5. E.g. 500, 25, 2795, 35090, 33000

Divisibility test by 4

A number is divisible by 4 when its last two digits are multiple of 4 i.e 00, 04, 08, 12, 16, 20, 24, 28, 32.....

Examples:

- a) Check whether 224 is divisible by 4
- 224 – The last two digits make 24 and 24 is divisible by 4
- Therefore 224 is divisible by 4

ACTIVITY:

- a) Which of the following numbers is divisible by 2
- a) 37 b) 9990 c) 179
- b) Test for divisibility by 3 and state whether the number is divisible by 3 or not
- 1. 63 c) 29631
- 2. 178
- c) Is 694 divisible by 4 or not?
- d) Check whether 3595 is divisible by 5
- e) Complete the table by using YES or NO

Number	By 2	By 3	By 4	By 5
3334	YES			
69250		NO		YES
1304	NO			
630001				
8896			YES	

REFERENCES:

- E.A.E.P Primary Maths book 6 page 16–17
- MK Maths Teachers’ book 6 page 76–77
- MK Maths Pupil’s book 6 page 65

LESSON :

TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : GEOMETRICAL SEQUENCES

CONTENT :

a) Square numbers

A square number is obtained or got by multiplying a number by itself.

$$\text{e.g. } 1^2 = 1 \times 1 = 1$$

$$2^2 = 2 \times 2 = 4$$

$$3^2 = 3 \times 3 = 9$$

$$4^2 = 4 \times 4 = 16$$

$$5^2 = 5 \times 5 = 25$$

$$6^2 = 6 \times 6 = 36$$

$$7^2 = 7 \times 7 = 49$$

$$8^2 = 8 \times 8 = 64$$

Therefore 1, 4, 9, 16, 25, 36, 49, 64, are square numbers

b) Triangular numbers

Triangular numbers are obtained by adding consecutive numbers. They can be represented as the pattern below.

$$1 = 1$$

$$1 + 2 = 3$$

$$1 + 2 + 3 = 6$$

$$1 + 2 + 3 + 4 = 10$$

The sequence;

$$1, 3, 6, 10, 15, 21, 28, 36, \dots$$

c) Rectangular numbers

$$1 \times 2 = 2$$

$$2 \times 3 = 6$$

$$3 \times 4 = 12$$

$$4 \times 5 = 20$$

The pattern for rectangular numbers is in a rectangular form.

d) Cubic numbers.

These are numbers got by multiplying the same number three times

$$1^3 = 1 \times 1 \times 1 = 1$$

$$2^3 = 2 \times 2 \times 2 = 8$$

$$3^3 = 3 \times 3 \times 3 = 27$$

$$4^3 = 4 \times 4 \times 4 = 64$$

$$5^3 = 5 \times 5 \times 5 = 125$$

ACTIVITY

1. Write down all the square numbers between 10 and 65
2. Find the sum of the second and fifth triangular numbers
3. Find the first four cubic numbers
4. Find the value of 6^3
5. What is the square number of 99?

REFERENCES:

E.A.E.P Primary Maths book 6 page 18–19

Functional Primary Maths Pupil's book 6 page 62–64

MK Maths Teachers' book 6 page 76–78

MK Maths Pupil's book 6 page 65–69

LESSON :

TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : ARITHMETIC PROGRESSION

CONTENT :

1. Even numbers

Numbers which are divisible by 2 with no remainder e.g. 0, 2, 4, 6, 8, 10, 12, 14, 16.....

2. Odd numbers

Numbers you divide by two and get a remainder as 1 e.g. 1, 3, 5, 7, 9, 11, 13, 15.....

Note; the pattern for an even and odd number is by adding 2

3. Counting numbers

These are numbers from 1 up to no end.

They are also called Natural numbers e.g. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,.....

4. Prime numbers

A prime number is a number with only two factors, which is 1 and itself, e.g. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.....

5. Composite numbers

A composite number is a number with more than two factors. E.g. 4, 6, 8, 9, 10, 12, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 29, 30.....

ACTIVITY

1. Find the sum of the first four even numbers
2. What is the product of the second and sixth odd number?
3. Divide the tenth counting number by the first prime number.
4. What is the difference between the sixth composite number and the third prime number?
5. Write all prime numbers between 20 and 36

REFERENCE

MK Maths Teachers' book 6 page 79-81

MK Maths Pupil's book 6 page 73-80

LESSON :

TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : CONSECUTIVE NUMBERS

CONTENT :

- i) The sum of three consecutive counting numbers is 15. Find the numbers.

Let the first number be Y

The pattern of counting numbers is by adding

$$1^{\text{st}} \text{ be } y; \quad 2^{\text{nd}} = y+1 \quad 3^{\text{rd}} = y+2$$

$$\text{Sum} = 15$$

$$y+y+1+y+2 = 15$$

$$y+y+y+1+2 = 15$$

$$3y+3 = 15$$

$$3y+3-3 = 15-3$$

$$3y/3 = 12/3$$

$$y = 4$$

$$y = 4$$

$$y+1 = 4+1 = 5$$

$$y+2 = 4+2 = 6$$

The numbers are 4, 5, 6

- ii) Find the sum of four consecutive even numbers when the smallest number is 6

$$6, 8, 10, 12$$

$$= 6+8+10+12 = 36$$

ACTIVITY

1. Musa wrote three consecutive counting numbers on the chalkboard. If the second number was 9, find the sum of the numbers he wrote.
2. The median of three consecutive odd numbers is 21. Find the numbers
3. Find the four consecutive counting numbers if their sum is 86

4. The total of three consecutive even numbers is 60. Find their range

REFERENCE

MK Maths Teachers' book 6 page 80–81

MK Maths Pupil's book 6 page 76–78

WEEK

LESSON: 1

TOPIC: NUMBER PATTERNS AND SEQUENCES

SUBTOPIC: PRIME FACTORISATION

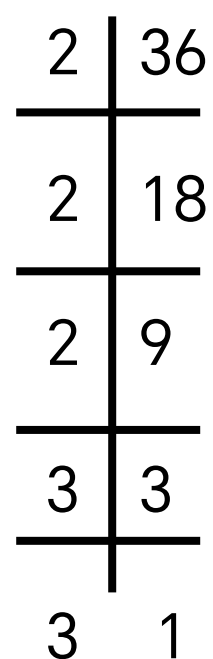
CONTENT:

- Prime factorization can be done in two methods.
 1. Using ladder method
 2. Using factortree method
- Prime factorization can also be in power form or subscript form.

Examples

a) Prime factorize 36 and give your answer in a power form.

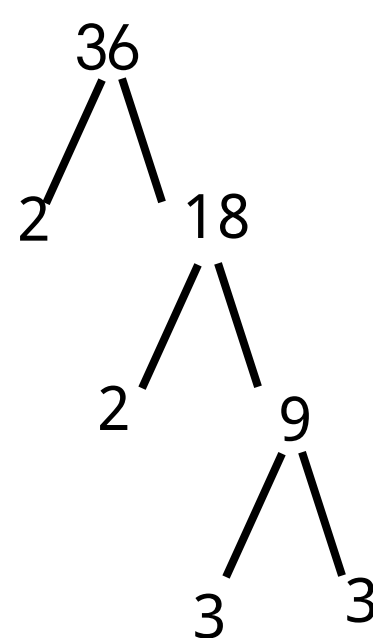
F₃₆



$$= (2 \times 2) \times (3 \times 3)$$

$$= 2^2 \times 3^2$$

F₃₆



$$= (2 \times 2) \times (3 \times 3)$$

$$= 2^2 \times 3^2$$

b) Prime factorize 48 and give your answer in subscript form

$$F_{48} = 2 \times 2 \times 2 \times 2 \times 3$$
$$= \{2_1, 2_2, 2_3, 2_4, 3_1\}$$

NOTE: Subscript form is also called set form.

ACTIVITY

a) Prime factorize the following and give the answer in a power form

- | | |
|-------|--------|
| 1. 12 | d) 100 |
| 2. 24 | e) 125 |
| 3. 72 | f) 18 |

b) Prime factorize and give the answer in subscript

- | | | | |
|-------|-------|-------|--------|
| 1. 90 | b) 32 | c) 15 | d) 120 |
|-------|-------|-------|--------|

REFERENCE

Functional Primary Maths Pupil's book 6 page 65–67

MK Maths Teachers' book 6 page 82–84

MK Maths Pupil's book 6 page 83–84

LESSON :

TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : FINDING PRIME FACTORISED NUMBERS

CONTENT :

1. What number has been expanded below?

$$2 \times 2 \times 3 \times 3 \times 3$$

$$= (2 \times 2) \times (3 \times 3) \times 3$$

$$= 4 \times 9 \times 3$$

$$= 36 \times 3$$

$$= 108$$

2. Find the prime factorized number to get $2^3 \times 3^2$

$$2^3 \times 3^2 = (2 \times 2 \times 2) \times (3 \times 3)$$

$$= 8 \times 9$$

$$= 72$$

3. Find the number that has been expanded below;

$$\{2_1, 2_2, 2_3, 3_1, 5_1\}$$

$$= 2_1 \times 2_2 \times 2_3 \times 3_1 \times 5_1$$

$$= (2 \times 2 \times 2) \times (3 \times 5)$$

$$= 8 \times 15$$

$$= 120$$

ACTIVITY

Find the numbers which have been prime factorized;

1. $2 \times 3 \times 5$

6. $\{2_1, 3_1, 5_1\}$

2. $2 \times 3 \times 3 \times 5$

7. $\{7_1, 11_1\}$

3. $2^3 \times 3^1$

8. $7^1 \times 11^1 \times 13^1$

4. $5^2 \times 7^1$

REFERENCES

MK Maths Teachers' book 6 page 82-84

Functional Primary Maths Pupil's book 6 page 67-68

LESSON :

TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : PRIME FACTORS ON VENN DIAGRAMS

CONTENT :

1. Prime factorize 18 and 12 and represent their prime factors on a venn diagram

Note: On a venn diagram, we put subscripts. So we prime factorize in subscript form.

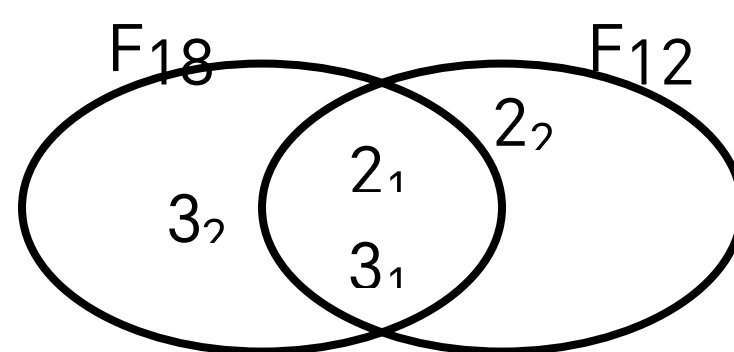
$$\begin{array}{r|l} 2 & 18 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline 1 & \end{array}$$

$$18 = 2_1 \ 2_2 \ 3_1 \ 3_2$$

$$\begin{array}{r|l} 2 & 12 \\ \hline 2 & 6 \\ \hline 3 & 3 \\ \hline \end{array}$$

$$12 = 2_1 \ 2_2 \ 2_3 \ 3_1$$

Common prime factors $\{2_1, 3_1\}$



Find the G.C.F of F_{18} and F_{12}

$$\begin{aligned} \text{G.C.F} &= \text{Product of factors in } F_{18} \cap F_{12}. \\ &= 2_1 \times 3_1 \\ &= 2 \times 3 \\ &= 6 \end{aligned}$$

2. Work out the L.C.M of F_{18} and F_{12}

L.C.M = Product of factors $F_{18} \cup F_{12}$.

$$\text{L.C.M} = 2_1 \times 3_1 \times 2_2 \times 3_2$$

$$\text{L.C.M} = 2 \times 3 \times 2 \times 3$$

$$= 6 \times 6$$

$$\text{L.C.M} = 36.$$

ACTIVITY

- a) Prime factorize 40 and 15 and represent their prime factors on a venn diagram

b) Find their L.C.M

c) Calculate their G.C.F

REFERENCES

Functional Primary Maths Pupil's book 6 page 68–69

MK Maths Pupil's book 6 page 82–85

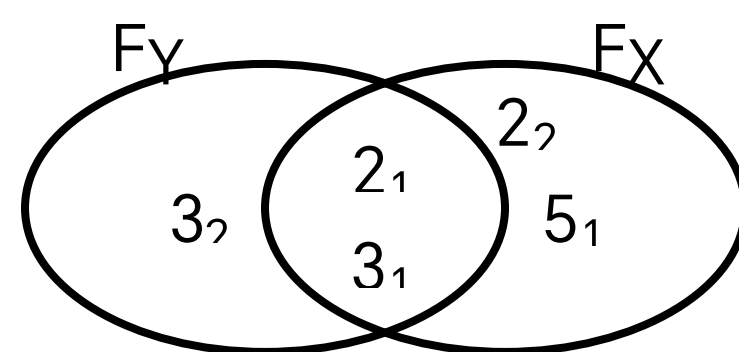
LESSON :

TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : INTERPRETING VENN DIAGRAMS

CONTENT :

1. The venn diagram below shows the prime factors of Y and X



a) Find the value of Y

$$Y = 2_1 \times 3_1 \times 3_2$$

$$Y = 2 \times 3 \times 3$$

$$Y = 6 \times 3$$

$$Y = 18.$$

b) Find the value of X

$$X = 2_1 \times 2_2 \times 3_1 \times 5_1$$

$$X = 2 \times 2 \times 3 \times 5$$

$$X = 12 \times 5$$

$$X = 60.$$

c) Find the L.C.M of F_Y and F_X

L.C.M = Product of factors $F_Y \cup F_X$.

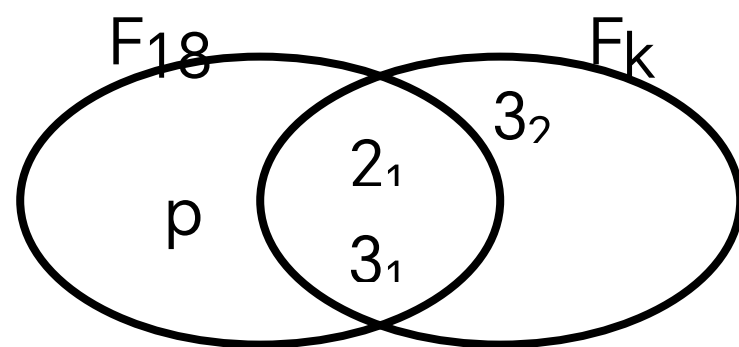
$$L.C.M = 2_1 \times 2_2 \times 5_1 \times 3_1 \times 3_2$$

$$= (2 \times 2 \times 5) \times (3 \times 3)$$

$$= 20 \times 9$$

$$\underline{L.C.M = 180}$$

2. a) Find the value of P and K in the figure



$$p \times 2_1 \times 3 = 12$$

$$p \times 2 \times 3 = 12$$

$$\underline{6P = 12}$$

$$6$$

$$6$$

$$\underline{P = 2.}$$

$$K = 2_1 \times 3_1 \times 3_2$$

$$K = 2 \times 3 \times 3$$

$$K = 6 \times 3$$

$$\underline{K = 18}$$

b) Find the G.C.F of F_{12} and F_k

G.C.F = Product of Intersection

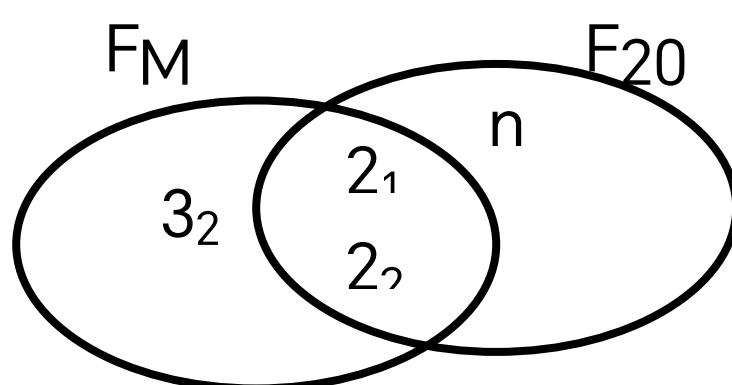
$$= 2_1 \times 3_1$$

$$= 2 \times 3$$

$$\underline{G.C.F = 6}$$

ACTIVITY

i) Given the venn diagram below



1. Find the value of
 - i) m
 - ii) n
2. Work out the L.C.M of F_m and F_{20}
3. Calculate the G.C.F of F_m and F_{20}

REFERENCES

Functional Primary Maths Pupil's book 6 page 68–70

LESSON :

TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : SQUARES AND SQUARE ROOTS OF WHOLE NUMBERS

CONTENT :

Squares:

1. Find the square of 5

$$5^2 = 5 \times 5 = 25$$

2. What is the square of 16

$$16^2 = 16 \times 16 = 256$$

Square Roots:

- a) Find the square root of 9

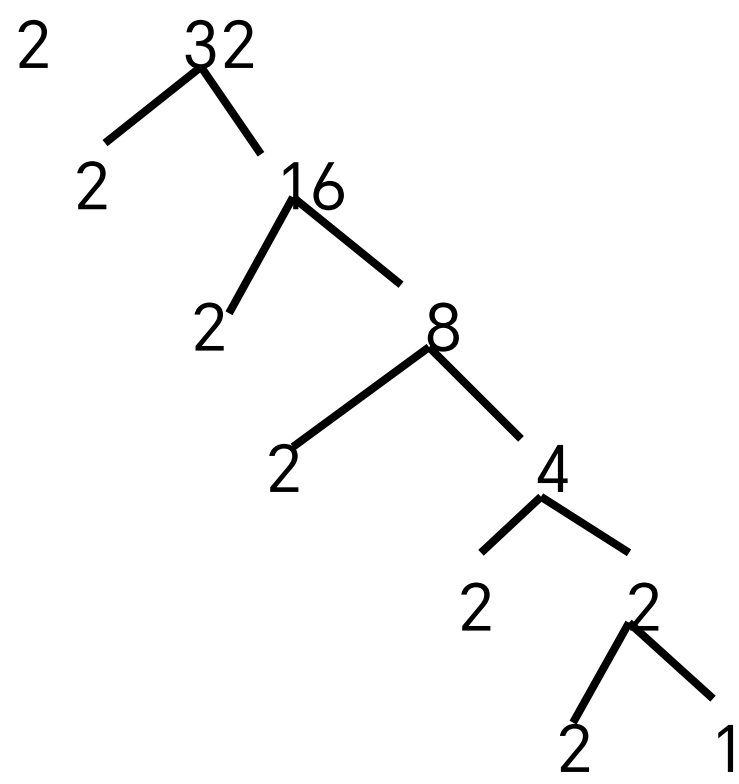
$$\begin{array}{r} \sqrt{9} = \begin{array}{r} 3 \overline{) 9} \\ \underline{3 } 3 \\ \underline{3 } 0 \end{array} = 3 \times 3 = 3^2 \end{array}$$

$$\sqrt{9} = \sqrt{3^2} = 3$$

Therefore $\sqrt{9} = 3$.

- b) Work out the square root of 64

$$\sqrt{64} = \sqrt{64}$$



$$= 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

$$= 2^6$$

$$\text{So, } \sqrt[2]{64} = \sqrt[2]{2^6} = 2^3 = 2 \times 2 \times 2 = 8$$

Therefore, $\sqrt{64} = 8$.

Square roots can also be got by using odd numbers

c) Find the square root of 6

$$16 - 1 = 15$$

$$15 - 3 = 12$$

$$12 - 5 = 7$$

$$7 - 7 = 0$$

Therefore, $\sqrt{16} = 4$.

Note: We subtract odd numbers in their order sequence until we get 0 and count the number of odd numbers used.

ACTIVITY:

a) Find the square root of each

1. 4

d) 121

2. 25

e) 196

3. 100

f) 255

b) Find the square of each

1. 6

c) 100

2. 10

d) 25

REFERENCE

MK Maths Teachers' book 6 page 88

MK Maths Pupil's book 6 page 95–97

Functional Primary Maths Pupil's book 6 page 71–74

LESSON :

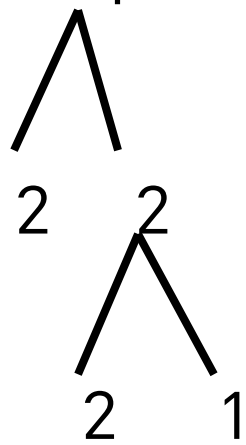
TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : SQUARE ROOTS OF FRACTIONS

CONTENT :

a) Find the square root of $\frac{4}{9}$

$$\frac{\sqrt{4}}{\sqrt{9}} = \frac{\sqrt{4}}{\sqrt{9}} = \frac{2}{3} \quad = \sqrt[2]{\frac{4}{9}} = \frac{2}{3}$$



/

$$\sqrt{\frac{4}{9}} = \frac{\sqrt{4}}{\sqrt{9}} = \frac{2}{3} \quad = \frac{2}{3} \quad = \sqrt[2]{\frac{4}{9}} = \frac{2}{3}$$

$$\text{Hence } \sqrt{\frac{4}{9}} = \frac{2}{3}$$

b) Work out the square root of $\frac{27}{9}$

$$\begin{aligned}
 \sqrt{27/9} &= \sqrt{25/9} = \frac{\sqrt{25}}{\sqrt{9}} & \begin{array}{r} 5 \ 25 \\ \sqrt{9} \overline{) 5 \ 25} \\ \underline{5 \ 25} \\ 1 \end{array} & \begin{array}{r} 3 \ 9 \\ \sqrt{9} \overline{) 3 \ 9} \\ \underline{3 \ 9} \\ 1 \end{array} \\
 &= \sqrt{5 \times 5} & & = \sqrt{3 \times 3} \\
 &= \sqrt{5^2} & & = \sqrt{3^2} \\
 &= \underline{5.} & & \underline{\underline{= 3.}}
 \end{aligned}$$

$$\begin{aligned}
 \text{SO } \sqrt{25/9} &= 5/3 \\
 &= \sqrt{27/9} = \underline{12/3.}
 \end{aligned}$$

ACTIVITY

a) Work out the square roots of the following

- | | | | |
|---------------|-------------|-------------|-----------|
| 1. $100/1000$ | c) $9/16$ | e) $61/4$ | g) $17/9$ |
| 2. $1/4$ | d) $119/81$ | f) $81/100$ | |

REFERENCE

MK Maths Teachers' book 6 page 88-89

MK Maths Pupil's book 6 page 98-100

MK Maths Pupil's book 7 page 56-57

Understanding Maths Pupil's book 7 page 43-44

LESSON :

TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : SQUARES ROOTS OF DECIMALS

CONTENT :

- Find the square root of 0.49

$$\frac{\sqrt{0.49}}{\sqrt{100}} = \frac{\sqrt{49}}{\sqrt{100}} = \frac{7}{10} = 0.7$$

$$\begin{array}{r} \sqrt{49} \quad \underline{7 \ 49} \\ 100 \quad \underline{7 \ 7} \\ 1 \end{array} \quad = \sqrt{7 \times 7} = \sqrt{7^2} = 7$$

$$\begin{array}{r} \sqrt{100} \quad \underline{2 \ 100} \\ \underline{2 \ 50} \\ \underline{5 \ 25} \\ \underline{5 \ 5} \\ 1 \end{array} \quad = \sqrt{2 \times 2 \times 5 \times 5} = \sqrt{2^2 \times 5^2} = 2 \times 5 = 10$$

Note: Square roots of decimals, we change the decimal into a common fraction first and after the square root of each part, we take it back to decimal form.

2. Work out the square root of 0.0081

$$\sqrt{0.0081} = \frac{\sqrt{81}}{\sqrt{10000}} = \frac{9}{100} = 0.09$$

ACTIVITY

Find the square root of each decimal;

- | | |
|---------|-----------|
| 1. 0.36 | 5) 2.25 |
| 2. 0.81 | 6) 1.21 |
| 3. 1.44 | 7) 0.0004 |
| 4. 1.96 | 8) 0.0064 |

REFERENCE

Understanding Maths Pupil's book 7 page 45-46

MK Maths Pupil's book 7 page

MK Maths Pupil's book 6 page 101

MK Maths Teachers' book 6 page 89

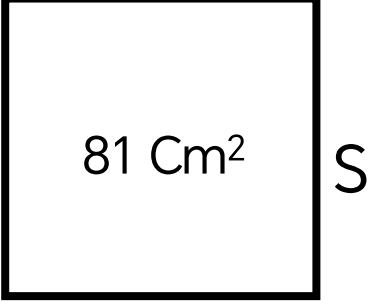
LESSON :

TOPIC : NUMBER PATTERNS AND SEQUENCES

SUBTOPIC : MORE ON SQUARES ROOTS

CONTENT :

1. The area of a square is 81 cm². Find the length of each side of the square.



S

$$\begin{aligned} \text{Area} &= S^2 \\ 81 \text{ cm}^2 &= S^2 \\ \sqrt{81 \text{ cm}^2} &= \sqrt{S^2} \\ S &= 9 \text{ cm} \end{aligned}$$

$$\begin{array}{r|l} 3 & 81 \\ \hline 3 & 27 \\ \hline 3 & 9 \\ \hline & \\ \hline & \end{array}$$

$$\begin{aligned} \sqrt{81} &= \sqrt{(3 \times 3 \times 3 \times 3 \times 3)} \\ &= \sqrt{3^4} \\ &= 3 \times 3 \\ &= 9 \end{aligned}$$

Therefore, Side = 9 Cm

2. Solve, $K^2 = 0.0004$

$$\sqrt{K^2} = \sqrt{0.0004}$$

$$\sqrt{K^2} = \sqrt{\frac{4}{10000}}$$

$$K = \frac{\sqrt{4}}{\sqrt{10000}}$$

$$K = 0.02$$

Solve $2Y^2 = 50$

$$2y^2 = 50$$

Dividing each term by the coefficient

$$\frac{2y^2}{2} = \frac{50}{2}$$

$$y^2 = 25$$

$$Y^2 = 25$$

Taking the square root on both sides

$$\sqrt{y^2} = \sqrt{25}$$

$$\underline{y} = \underline{5}$$

$$\begin{array}{r|l} & 5 \ 25 \\ \hline 2 \ 1 & \end{array}$$

$$\begin{aligned} \sqrt{25} &= 5 \times 5 \\ &= \sqrt{5^2} \end{aligned}$$

ACTIVITY

Mulindwa's square garden covers an area of 100 m^2 . Calculate the length of each side of the garden.

The base area of a cube is 16 cm^2 . Find its height.

A farmer used a barbed wire to fence his square garden that covers an area of 196 m^2 . Find the length of the barbed wire used.

Solve $P^2 = 1$

Solve $900 = m^2$

Find the value of $3r^2 = 12$

REFERENCES

LESSON :

TOPIC : FRACTIONS

SUBTOPIC : ADDITION OF FRACTIONS

CONTENT :

1. Add: $\frac{5}{6} + \frac{1}{4}$

$$\frac{5}{6} + \frac{1}{4}$$

6 4 LCM 12

$$= \frac{10 + 3}{12}$$

12

$$= 1 \frac{3}{12}$$

$$= \underline{1 \frac{1}{4}}$$

$$\begin{array}{r} 5. \quad 12 \\ \hline 3. \quad 6 \\ \hline 1. \quad 3 \\ \hline 1 \end{array}$$

$$\text{LCM} = 2 \times 2 \times 3 = 12$$

Method ii

$$\frac{5}{6} + \frac{1}{4}$$

$$= \frac{20 + 6}{24}$$

24

$$= \frac{26}{24}$$

$$= \underline{1 \frac{1}{12}}$$

$$\begin{array}{r} 5 \times 4 + 1 \times 6 \\ \hline 6 \times 4 \quad 4 \times 6 \end{array}$$

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$$= \frac{20}{24} + \frac{6}{24}$$

2. Find the sum of $\frac{1}{2} + \frac{1}{4} + \frac{1}{3}$

$$= \frac{6 + 3 + 4}{12}$$

$$= \frac{13}{12} = 1\frac{1}{12}$$

ACTIVITY

Add the following fractions

$$\frac{10}{6} + \frac{5}{4} + \frac{1}{3} \qquad \frac{4}{6} + \frac{4}{4} + \frac{2}{3}$$

$$\frac{11}{6} + \frac{5}{4} + \frac{15}{3} + \frac{1}{4}$$

$$\frac{12}{6} + \frac{5}{4} + \frac{1}{3} \qquad \frac{6}{4} + \frac{2}{4} + \frac{1}{6}$$

Lule has $4\frac{1}{4}$ meters of cloth and Sam has $9\frac{1}{6}$ meters. Find their total length.

REFERENCES

Functional Primary Maths Pupil's book 6 page 91-92

MK Maths Teachers' book 6 page 98-99

MK Maths Pupil's book 6 page 106

LESSON :

TOPIC : FRACTIONS

SUBTOPIC :SUBTRACTION OF FRACTIONS

CONTENT :

1. Subtract: $\frac{4}{5} - \frac{1}{2}$
$$\frac{4}{5} - \frac{1}{2} = \frac{8}{10} - \frac{5}{10} = \frac{3}{10}$$
2. Subtract $\frac{2}{3}$ from $4\frac{1}{4}$
$$= 4\frac{1}{4} - \frac{2}{3}$$

ACTIVITY

Subtract the following fractions

1. $\frac{4}{5} - \frac{1}{2}$

$$\frac{4}{5} - \frac{1}{2} = \frac{8}{10} - \frac{5}{10} = \frac{3}{10}$$
2. $\frac{4}{5} - \frac{1}{2}$

$$\frac{4}{5} - \frac{1}{2} = \frac{8}{10} - \frac{5}{10} = \frac{3}{10}$$
3. $\frac{4}{5} - \frac{1}{2}$

$$\frac{4}{5} - \frac{1}{2} = \frac{8}{10} - \frac{5}{10} = \frac{3}{10}$$

7. Annette was given $4\frac{2}{3}$ Kg of sugar and gave out $2\frac{1}{2}$ kg. What fraction did she remain with?

$$4\frac{2}{3} - 2\frac{1}{2} = \frac{10}{3} - \frac{5}{2} = \frac{20}{6} - \frac{15}{6} = \frac{5}{6}$$

8. Kato had 6 trays of eggs. He used $1\frac{1}{3}$ trays. What fraction remained?

$$6 - 1\frac{1}{3} = \frac{12}{3} - \frac{4}{3} = \frac{8}{3}$$

REFERENCES

- Functional Primary Maths Pupil’s book 6 page 93–94
- MK Maths Teachers’ book 6 page 98–99
- MK Maths Pupil’s book 6 page 106

LESSON :

TOPIC : FRACTIONS

SUBTOPIC : MULTIPLICATION OF FRACTIONS

CONTENT :

1. Multiply: $\frac{2}{3} \times \frac{1}{5}$

$$\frac{2 \times 1}{3 \times 5} = \frac{2}{15}$$

2. Find the product of $\frac{3}{5}$ and $\frac{1}{4}$

$$\frac{3}{4} \times \frac{1}{5}$$

$$\frac{3 \times 1}{4 \times 5} = \frac{3}{20} = 1 \frac{3}{10}$$

3. What is $\frac{1}{4}$ of $\frac{1}{3}$

1. $\frac{1}{4}$

$$\frac{1}{4} \text{ of } \frac{1}{3} = \frac{1}{4} \times \frac{1}{3} = \frac{1 \times 1}{4 \times 3} = \frac{1}{12}$$

ACTIVITY

1. Multiply the following fractions

a) $\frac{1}{6} \times \frac{4}{9}$

c) $\frac{3}{5} \times \frac{2}{7}$

b) $\frac{17}{3} \times \frac{1}{2}$

d) $\frac{1}{2} \times \frac{1}{3} \times \frac{1}{4}$

2. Find the product of $\frac{2}{9}$ and $\frac{1}{8}$

3. What is $\frac{2}{5}$ of 90?

4. What is the value $\frac{1}{2}$ of $\frac{1}{2}$

REFERENCES

Functional Primary Maths Pupil's Book 6 page 79

MK Maths Teachers' Book 6 page 99–104

MK Maths Pupil's Book 6 page 106–108

LESSON :

TOPIC : FRACTIONS

SUBTOPIC : DIVISION OF FRACTIONS

CONTENT :

: Reciprocal

Note; A reciprocal is a number multiplied by the given number to give a product 1.

e.g. What is the reciprocal of 4?

Let the reciprocal be m

So $m \times 4 = 1$

$$\frac{4m}{4} = \frac{1}{1} = m \frac{1}{4}$$

Therefore the reciprocal of 4 is $= \frac{1}{4}$

In short, the reciprocal of any fraction is the denominator to become a numerator and vice versa.

Examples:

a) Work out: $\frac{1}{2} \div \frac{1}{4}$

Note: When dividing fractions,

- The first fraction remains.
- Division sign change to multiplication
- The second fraction changes to its reciprocal

$$\text{So, } \frac{1}{2} \div \frac{1}{4} = \frac{1}{2} \times \frac{4}{1} = \frac{1 \times 4}{2 \times 1} = \frac{4}{2} = 2$$

$$\text{Divide: } \frac{2}{5} \div \frac{1}{3}$$

$$\left(\frac{2 \times 5 \times 3}{1^{1/5}} \right) \div \left(\frac{1 \times 5 \times 3}{5} \right) = \frac{2 \times 3 \div 1 \times 5}{3} = \frac{6}{5} \div 5 = 6 =$$

We multiply each fraction by product of the denominators

$$\text{b) Divide: } \frac{4}{9} \div \frac{1}{3}$$

$$\left(\frac{2 \times 5 \times 3}{1} \right) \div \left(\frac{1 \times 5 \times 3}{5} \right) = \frac{2 \times 3 \div 1 \times 5}{3} = \frac{6}{5} \div 5 = 6 = 1$$

ACTIVITY

1. Divide the following fractions

$$1. \frac{4}{9} \div \frac{1}{9} \quad \text{d) } \frac{1}{3} \div \frac{1}{3} \quad \text{g) } \frac{6}{3} \div \frac{1}{3} \quad 3$$

$$2. \frac{4}{9} \div \frac{1}{9} \quad \text{e) } \frac{6}{3} \div \frac{1}{3} \quad \text{h) } \frac{1}{8} \div \frac{10}{3} \quad 5$$

$$3. \frac{4}{3} \div \frac{1}{3} \quad \text{f) } \frac{10}{15} \div \frac{2}{3} \quad 9$$

REFERENCES

MK Maths Teachers' book 6 page 107-111

MK Maths Pupil's book 6 page 1111-112

Functional Primary Maths Pupil's book 6 page 80-87

LESSON :

TOPIC : FRACTIONS

SUBTOPIC : MIXED OPERATIONS IN FRACTIONS

CONTENT :

In mixed operations, we use “**BODMAS**” to handle the operations

B – Brackets

M – Multiplication

O – Of

A – Addition

D – Division

S – Subtraction

a) Workout: $\frac{1}{2} + \frac{1}{4} \times \frac{1}{3}$ of $\frac{1}{5}$

BODMAS

$$\frac{1}{2} + \frac{1}{4} \times \left(\frac{1}{3} \text{ of } \frac{1}{5} \right)$$

$$\frac{1}{2} + \frac{1}{4} \times \left(\frac{1}{3} \times \frac{1}{5} \right)$$

$$\frac{1}{2} + \frac{1}{4} \times \frac{1 \times 1}{3 \times 5}$$

$$\frac{1}{2} + \frac{1}{4} \times \frac{1}{15}$$

$$\frac{1}{2} + \left(\frac{1}{4} \times \frac{1}{15} \right)$$

$$\frac{1}{2} + \frac{1}{4} \times \frac{1}{15} = \frac{1}{2} + \frac{1}{60} = \frac{30}{60} + \frac{1}{60} = \frac{31}{60}$$

b) Simplify: $\frac{4}{5} - \frac{1}{2} + \frac{1}{3}$

BODMAS

$$\left(\frac{4}{5} + \frac{1}{3} \right) - \frac{1}{2} = \left(\frac{12}{15} + \frac{5}{15} \right) - \frac{1}{2} = \frac{17}{15} - \frac{1}{2} = \frac{34}{30} - \frac{15}{30} = \frac{19}{30}$$

ACTIVITY:

Work out the following

$$\frac{4}{2} - \frac{1}{4} \times \frac{1}{3} + \frac{1}{5}$$

$$\frac{5}{7} - \frac{4}{2} \text{ of } \frac{1}{9} + \frac{4}{5}$$

$$\frac{6}{5} - \frac{2}{9} \div \frac{6}{3} \times \frac{1}{2} \left(\frac{1}{5} - \frac{1}{3} \right)$$

1. $\frac{2}{5} - \frac{1}{4} + \frac{6}{9}$

6. $\frac{4}{10} - \frac{1}{2} + \frac{1}{2}$

$$\frac{7}{5} - \frac{9}{3} + \frac{1}{2} - \frac{1}{5} \times 3 \div \frac{1}{4}$$

REFERENCES

MK Maths Teachers' Book 6 Page 112-114

MK Maths Pupil's Book 6 Page 113-114

LESSON :

TOPIC : FRACTIONS

SUBTOPIC : ADDITION AND SUBTRACTION OF DECIMALS

CONTENT :

1. Add: $3.4 + 24.63$

$$\begin{array}{r} 24.63 \\ + 3.40 \\ \hline 28.03 \end{array}$$

2. Alex had 7.05 meters and Kato had 17.13 meters of string. Find the total length of their string. Find the total length of their strings.

$$\begin{array}{r} 17.13 \text{ metres} \\ + 7.05 \text{ metres} \\ \hline 24.18 \text{ metres} \end{array}$$

3. Subtract $9.5 - 3.6$

$$\begin{array}{r} 9.5 \\ - 3.6 \\ \hline 5.9 \end{array}$$

4. Subtract 0.9 from 100

$$\begin{array}{r} 100.0 \\ - 0.9 \\ \hline 99.1 \end{array}$$

ACTIVITY:

1. Work out the following

1. $6.4 + 9.6$ d) $76.6 - 67.7$
2. $166.66 + 0.4$ e) $100 - 0.101$
3. $5.55 + 555$ f) $30 - 0.3$

2. Subtract 0.2 from 20

3. Subtract: $1000 - 0.01$

REFERENCES

Functional Primary Maths Pupil's book 6 page 94-95

MK Maths Teachers' book 6 page 113-116

MK Maths Pupil's book 6 page 114-115

LESSON :

TOPIC : FRACTIONS

SUBTOPIC : MULTIPLICATION OF DECIMALS

CONTENT :

1. Multiply: 0.3×6

$$\begin{array}{r}
 0.3 \\
 \times 6 \\
 \hline
 1.8
 \end{array}
 \qquad
 \begin{array}{l}
 6 \times 3 = 18 \\
 6 \times 0 = 0 + 1 = 1
 \end{array}$$

2. Multiply: 4.5×2.6

$$\begin{array}{r}
 45 \times 26 \\
 10 \quad 10 \\
 \hline
 1170 \\
 100 \quad = 11.7
 \end{array}$$

3. Find the product of 2.34 and 1.2

$$\begin{array}{r} 2.34 \\ \times 1.2 \\ \hline \end{array}$$

4168

$$\begin{array}{r} +2340 \\ \hline \end{array}$$

2.808

After multiplying and adding, we consider the decimal places altogether.

ACTIVITY:

Work out the following

1 3.3×2

5) 32.5×0.3

2 6.6×1.2

6) 9.8×2.2

3 9.4×100

7) 144.4×100

4 17.17×2.5

8) 33×3.3

REFERENCES

MK Maths Teachers' book 6 page 117-119

MK Maths Pupil's book 6 page 116-119

LESSON :

TOPIC : FRACTIONS

SUBTOPIC : DIVISION OF DECIMALS

CONTENT :

1. Divide 6 by 0.2

$$6 \div 0.2$$

$$6 \div \underline{2}$$

10

$$\begin{array}{r} \underline{6} \times \underline{10} \\ 1 \quad 2 \end{array}$$

1. Divide 0.5 by 20

$$0.5 \div 20$$

$$\begin{array}{r} 5 \div 20 \\ 10 \quad 1 \end{array}$$

$$\begin{array}{r} 5 \times 1 \\ 10 \quad 20 \end{array}$$

$$1 \times 1$$

$$10 \times 4$$

1.

40

$$\frac{3 \times 10}{1 \times 1}$$

$$\frac{30}{1}$$

$$= 30$$

2. Work out: $4.5 \div 2.5$

$$\frac{45}{10} \div \frac{25}{10}$$

$$\frac{45}{10} \times \frac{10}{25}$$

$$\frac{45}{25}$$

$$\frac{9}{5}$$

$$= 1.8$$

ACTIVITY

Work out the following

a) $12 \div 0.2$

5) $0.8 \div 2$

b) $8.5 \div 0.5$

6) $100 \div 0.1$

c) $14.4 \div 1.2$

7) $7.2 \div 0.06$

d) $0.96 \div 0.008$

8) $200 \div 0.002$

REFERENCES

Functional Primary Maths Pupil's book 6 page 98-99

MK Maths Teachers' book 6 page 119-120

MK Maths Pupil's book 6 page 118-119

LESSON :

TOPIC : FRACTIONS

SUBTOPIC : MULTIPLICATION AND DIVISION OF DECIMALS

CONTENT :

1) Work out: $\underline{1.2 \times 2.4}$

0.3

$$\left[\begin{array}{r} 12 \times 24 \\ \hline 10 \quad 10 \end{array} \right] \div \frac{3}{10}$$

$$\underline{12} \times \underline{24} \times \underline{10}$$

a) 10 3

$$\begin{array}{r} 6 \times 8 \times 1 \\ \hline 5 \times 1 \times 1 \end{array} \quad \begin{array}{r} 48 \\ \hline 5 \end{array} = 9.6$$

2) Simplify: $\underline{1.8 \times 7.2}$

0.2X0.03

$$\left[\begin{array}{r} 18 \times 72 \\ \hline 10 \quad 10 \end{array} \right] \div \left[\begin{array}{r} 2 \times 3 \\ \hline 10 \quad 100 \end{array} \right]$$

$$\frac{18 \times 72 \times 10 \times 100}{10 \quad 10 \quad 2 \quad 3} = \frac{3 \times 72 \times 10}{1 \times 1 \times 1 \times 1}$$

$$\underline{= 2160}$$

ACTIVITY

Work out the following:

1) $\underline{1.6 \times 7.2}$

0.06

4) 0.045×0.9

$$\underline{0.9 \times 0.03}$$

5) 6.4×2.8

$$\underline{0.04}$$

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6) 1.8×3.5

$$0.7 \times 0.3$$

2.) $\frac{12.3 \times 2.4}{}$

3.6

3) $\frac{36 \times 0.4}{}$

0.9

REFERENCES

Functional Primary Maths Pupil's book 6 page 99–100

MK Maths Teachers' book 6 page 122–123

MK Maths Pupil's book 6 page 121–122